

3. For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

c) $\{(2, 4), (4, 2)\}$

d) $\{(1, 2), (2, 3), (3, 4)\}$

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

6. Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x + y = 0$.

b) $x = \pm y$.

c) $x - y$ is a rational number.

d) $x = 2y$.

e) $xy \geq 0$.

f) $xy = 0$.

g) $x = 1$.

h) $x = 1$ or $y = 1$.

8. Show that the relation $R = \emptyset$ on a nonempty set S is symmetric and transitive, but not reflexive.

30. Let $R_1 = \{(1, 2), (2, 3), (3, 4)\}$ and $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$ be relations from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$. Find

a) $R_1 \cup R_2.$

c) $R_1 - R_2.$

b) $R_1 \cap R_2.$

d) $R_2 - R_1.$

32. Let R be the relation $\{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$, and let S be the relation $\{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$.

44. List the 16 different relations on the set $\{0, 1\}$.

47. a) How many relations are there on the set $\{a, b, c, d\}$?
- b) How many relations are there on the set $\{a, b, c, d\}$ that contain the pair (a, a) ?

22. Draw the directed graph that represents the relation $\{(a, a), (a, b), (b, c), (c, b), (c, d), (d, a), (d, b)\}$.

In Exercises 23–28 list the ordered pairs in the relations represented by the directed graphs.

30. How can the directed graph of a relation R on a finite set A be used to determine whether a relation is irreflexive?

2. Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers. What is the reflexive closure of R ?

4. How can the directed graph representing the reflexive closure of a relation on a finite set be constructed from the directed graph of the relation?

In Exercises 5–7 draw the directed graph of the reflexive closure of the relations with the directed graph shown.

24. Suppose that the relation R is irreflexive. Is the relation R^2 necessarily irreflexive?

1. Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

- a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
- b) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
- c) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
- d) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
- e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

2. Which of these relations on the set of all people are equivalence relations? Determine the properties of an equivalence relation that the others lack.
- a) $\{(a, b) \mid a \text{ and } b \text{ are the same age}\}$
 - b) $\{(a, b) \mid a \text{ and } b \text{ have the same parents}\}$
 - c) $\{(a, b) \mid a \text{ and } b \text{ share a common parent}\}$
 - d) $\{(a, b) \mid a \text{ and } b \text{ have met}\}$
 - e) $\{(a, b) \mid a \text{ and } b \text{ speak a common language}\}$

10. Suppose that A is a nonempty set and R is an equivalence relation on A . Show that there is a function f with A as its domain such that $(x, y) \in R$ if and only if $f(x) = f(y)$.

24. Determine whether the relations represented by these zero-one matrices are equivalence relations.

a)
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

26. What are the equivalence classes of the equivalence relations in Exercise 1?

2. Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (2, 2), (3, 3)\}$

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 1), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (2, 3), (3, 0), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (3, 3)\}$

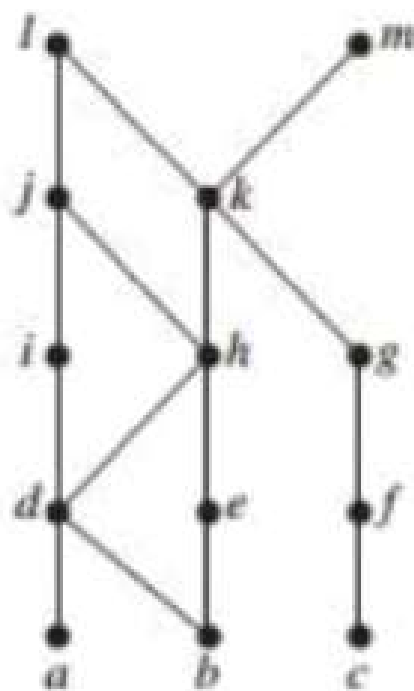
20. Draw the Hasse diagram for the greater than or equal to relation on $\{0, 1, 2, 3, 4, 5\}$.

22. Draw the Hasse diagram for divisibility on the set

- a) $\{1, 2, 3, 4, 5, 6\}$. b) $\{3, 5, 7, 11, 13, 16, 17\}$.
c) $\{2, 3, 5, 10, 11, 15, 25\}$. d) $\{1, 3, 9, 27, 81, 243\}$.

24. Draw the Hasse diagram for inclusion on the set $P(S)$, where $S = \{a, b, c, d\}$.

32. Answer these questions for the partial order represented by this Hasse diagram.



- Find the maximal elements.
- Find the minimal elements.
- Is there a greatest element?
- Is there a least element?
- Find all upper bounds of $\{a, b, c\}$.
- Find the least upper bound of $\{a, b, c\}$, if it exists.
- Find all lower bounds of $\{f, g, h\}$.
- Find the greatest lower bound of $\{f, g, h\}$, if it exists.

62. Find a compatible total order for the divisibility relation on the set $\{1, 2, 3, 6, 8, 12, 24, 36\}$.